

Lecture 04 - Geometric Projection

This lecture explains how a 3D view becomes a 2D image. Projection is where camera geometry, homogeneous coordinates, clipping planes, depth precision, and viewport mapping meet.

Source chunk	course_text_parts/03_lectures/04_geometric-projection.txt
Extracted slide pages	61
Reader purpose	Commentary and causal explanation for human reading

How to read this lecture

- Read the section commentary before looking at the raw slide bullets.
- Use the mental model as the short version you should remember.
- Answer the check question without notes before continuing.

Slide walkthrough

Each row gives a short reading comment for one extracted slide page. Use it to turn sparse slide bullets into a human-readable path through the lecture.

Page	Slide title	How to read it
1	Untitled slide	Read this slide by connecting the bullet terms causally: what problem is being solved, what data is used, and what output is produced?
2	Projecting 3D Models	Read this as a coordinate-space step. Name the input space, the matrix or operation, and the output space.
3	4.1 Linear Perspective	Read this as pipeline state and data flow. Ask which OpenGL object or shader stage owns the data at this point.
4	4.1 Linear Perspective	Read this as camera geometry. Track how view-space positions become clip, normalized, and screen coordinates.
5	Linear Perspective	Read this as camera geometry. Track how view-space positions become clip, normalized, and screen coordinates.
6	Drawing by Projecting	Read this as camera geometry. Track how view-space positions become clip, normalized, and screen coordinates.
7	Projection in the Arts 1/5	Read this as camera geometry. Track how view-space positions become clip, normalized, and screen coordinates.
8	Projection in the Arts 2/5	Read this as camera geometry. Track how view-space positions become clip, normalized, and screen coordinates.

Page	Slide title	How to read it
9	Projection in the Arts 3/5	Read this as camera geometry. Track how view-space positions become clip, normalized, and screen coordinates.
10	Projection in the Arts 4/5	Read this as camera geometry. Track how view-space positions become clip, normalized, and screen coordinates.
11	Projection in the Arts 4/5	Read this as camera geometry. Track how view-space positions become clip, normalized, and screen coordinates.
12	Projection in the Arts 5/5	Read this as camera geometry. Track how view-space positions become clip, normalized, and screen coordinates.
13	Projection in the Arts 6/5	Read this as camera geometry. Track how view-space positions become clip, normalized, and screen coordinates.
14	Projection Technics 1/3	Read this as camera geometry. Track how view-space positions become clip, normalized, and screen coordinates.
15	Projection Technics 1/3	Read this as camera geometry. Track how view-space positions become clip, normalized, and screen coordinates.
16	Projection Technics 2/3	Read this as camera geometry. Track how view-space positions become clip, normalized, and screen coordinates.
17	Projection Technics 3/3	Read this as camera geometry. Track how view-space positions become clip, normalized, and screen coordinates.
18	Projection Technics 3/3	Read this as camera geometry. Track how view-space positions become clip, normalized, and screen coordinates.
19	4.2 Planar Projections	Read this as camera geometry. Track how view-space positions become clip, normalized, and screen coordinates.
20	Geometric Projections	Read this as camera geometry. Track how view-space positions become clip, normalized, and screen coordinates.
21	Projection Types 1/2	Read this as camera geometry. Track how view-space positions become clip, normalized, and screen coordinates.
22	Projection Types 2/2	Read this as camera geometry. Track how view-space positions become clip, normalized, and screen coordinates.
23	Orthographic Projections	Read this as camera geometry. Track how view-space positions become clip, normalized, and screen coordinates.

Page	Slide title	How to read it
24	Axometric Projections	Read this as a coordinate-space step. Name the input space, the matrix or operation, and the output space.
25	Isometric Projections	Read this as camera geometry. Track how view-space positions become clip, normalized, and screen coordinates.
26	Oblique Projection 1/2	Read this as camera geometry. Track how view-space positions become clip, normalized, and screen coordinates.
27	Oblique Projection 2/2	Read this as camera geometry. Track how view-space positions become clip, normalized, and screen coordinates.
28	Properties of Parallel Projections	Read this as camera geometry. Track how view-space positions become clip, normalized, and screen coordinates.
29	Perspective Projections 1/3	Read this as camera geometry. Track how view-space positions become clip, normalized, and screen coordinates.
30	Perspective Projections 2/3	Read this as a coordinate-space step. Name the input space, the matrix or operation, and the output space.
31	Perspective Projections 3/3	Read this as a coordinate-space step. Name the input space, the matrix or operation, and the output space.
32	4.3 Camera Modeling	Read this as camera geometry. Track how view-space positions become clip, normalized, and screen coordinates.
33	Camera Modeling	Read this as camera geometry. Track how view-space positions become clip, normalized, and screen coordinates.
34	View Volume	Read this as camera geometry. Track how view-space positions become clip, normalized, and screen coordinates.
35	Projection Specification	Read this as camera geometry. Track how view-space positions become clip, normalized, and screen coordinates.
36	View Angle	Read this as camera geometry. Track how view-space positions become clip, normalized, and screen coordinates.
37	Clipping Planes	Read this as camera geometry. Track how view-space positions become clip, normalized, and screen coordinates.
38	4.4 Specifying Projections in OpenGL	Read this as pipeline state and data flow. Ask which OpenGL object or shader stage owns the data at this point.

Page	Slide title	How to read it
39	Projection Matrix Generation 1/3	Read this as a coordinate-space step. Name the input space, the matrix or operation, and the output space.
40	Projection Matrix Generation 2/3	Read this as a coordinate-space step. Name the input space, the matrix or operation, and the output space.
41	Projection Matrix Generation 3/3	Read this as a coordinate-space step. Name the input space, the matrix or operation, and the output space.
42	Orthographic Matrix Specification	Read this as a coordinate-space step. Name the input space, the matrix or operation, and the output space.
43	4.5 Orthographic Projection Derivation	Read this as a coordinate-space step. Name the input space, the matrix or operation, and the output space.
44	Canonical View Volume	Read this as a coordinate-space step. Name the input space, the matrix or operation, and the output space.
45	Orthographic Projection 1/4	Read this as a coordinate-space step. Name the input space, the matrix or operation, and the output space.
46	Orthographic Projection 2/4	Read this as a coordinate-space step. Name the input space, the matrix or operation, and the output space.
47	Orthographic Projection 3/4	Read this as camera geometry. Track how view-space positions become clip, normalized, and screen coordinates.
48	Orthographic Projection 4/4	Read this as camera geometry. Track how view-space positions become clip, normalized, and screen coordinates.
49	4.6 Perspective Projection Derivation	Read this as a coordinate-space step. Name the input space, the matrix or operation, and the output space.
50	Perspective Projection	Read this as a coordinate-space step. Name the input space, the matrix or operation, and the output space.
51	Transformation of z-Values 1/2	Read this as a coordinate-space step. Name the input space, the matrix or operation, and the output space.
52	Transformation of z-Values 2/2	Read this as a coordinate-space step. Name the input space, the matrix or operation, and the output space.
53	Transformation of x- and y-Values 1/2	Read this as a coordinate-space step. Name the input space, the matrix or operation, and the output space.

Page	Slide title	How to read it
54	Transformation of x- and y-Values 2/2	Read this as a coordinate-space step. Name the input space, the matrix or operation, and the output space.
55	4.7 Viewport Transformation	Read this as a coordinate-space step. Name the input space, the matrix or operation, and the output space.
56	Screen Coordinate Mapping	Read this as a coordinate-space step. Name the input space, the matrix or operation, and the output space.
57	Viewport Transformation 1/2	Read this as a coordinate-space step. Name the input space, the matrix or operation, and the output space.
58	Viewport Transformation 2/2	Read this as a coordinate-space step. Name the input space, the matrix or operation, and the output space.
59	volume, we can differentiate different projection types	Read this as pipeline state and data flow. Ask which OpenGL object or shader stage owns the data at this point.
60	Literature and other sources used in this chapter	Read this slide by connecting the bullet terms causally: what problem is being solved, what data is used, and what output is produced?
61	- Text Books	Read this slide by connecting the bullet terms causally: what problem is being solved, what data is used, and what output is produced?

04.1 Linear Perspective and Planar Projections

Source location: Sections 4.1-4.2

Commentary

Perspective projection models the visual effect that farther objects appear smaller. Orthographic projection removes this distance-based size change and keeps parallel lines parallel. Both are useful, but they communicate different spatial relationships.

Planar projection means mapping points onto an image plane. The lecture's art examples are not decoration; they show that projection is a geometric rule for constructing an image from a viewpoint.

Mental model

Projection is the rule that turns 3D positions into image-plane positions.

Check yourself

What visual cue does perspective projection add that orthographic projection lacks?

04.2 Camera Modeling

Source location: Section 4.3

Commentary

A virtual camera is defined by position, orientation, projection type, viewing volume, and image/viewport settings. The view transform places the world relative to the camera. The projection transform maps that view into clip coordinates.

Near and far clipping planes are part of the camera model. They decide what range of depths is represented and strongly affect depth buffer precision. A careless near/far setup can create z-fighting even if the scene geometry is correct.

Mental model

A camera is not only an eye position; it is also a volume and a mapping rule.

Check yourself

Why can changing the near plane affect depth artifacts?

04.3 Specifying Projections in OpenGL

Source location: Section 4.4

Commentary

In OpenGL, projection is usually encoded in a projection matrix sent to a shader. The matrix maps view-space coordinates to clip space. Clipping and the perspective divide then lead toward normalized device coordinates.

Aspect ratio and field of view must match the intended viewport. If the aspect ratio is wrong, objects appear stretched. If the field of view is extreme, the image can look distorted even though the math is functioning.

Mental model

The projection matrix is the camera lens encoded as linear algebra in homogeneous coordinates.

Check yourself

What symptom suggests that the projection aspect ratio does not match the window?

04.4 Orthographic and Perspective Derivations

Source location: Sections 4.5-4.6

Commentary

The derivations are there to explain where the matrix entries come from. Orthographic projection maps a box-shaped view volume into normalized coordinates. Perspective projection maps a frustum and uses the homogeneous component for the later divide.

You do not need to treat these matrices as random formulas. Each part maps a coordinate range into a standard range. The perspective matrix additionally arranges w so that the perspective divide creates foreshortening.

Mental model

Projection matrices normalize a viewing volume; perspective also prepares the divide by w .

Check yourself

What does the perspective divide do to x and y coordinates?

04.5 Viewport Transformation

Source location: Section 4.7

Commentary

After normalized device coordinates, the viewport transform maps the normalized range to actual window coordinates. This is the last geometric mapping before rasterization uses the target grid.

Separating projection from viewport mapping helps debugging. A wrong projection can produce wrong normalized coordinates; a wrong viewport can map otherwise valid coordinates into the wrong part of the window.

Mental model

Projection decides normalized position; viewport decides where that position lands on the screen.

Check yourself

Why is resizing a window related to both viewport and projection settings?

End-of-lecture summary

If you remember only one thing from Lecture 04, remember this: This lecture explains how a 3D view becomes a 2D image. Projection is where camera geometry, homogeneous coordinates, clipping planes, depth precision, and viewport mapping meet.

Before moving on, you should be able to explain the lecture without reading slide bullets aloud. Use the vocabulary from the slides, but add causal links: why a step exists, what data it consumes, what it produces, and what can go wrong.